Carson Secrest

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Education

Oregon State University **Expected December 2026**
Bachelor of Science in Computer Science - **GPA: 3.97** *Corvallis, OR*

- **Relevant Coursework:** Algorithms, Data Structures, Object-Oriented Programming, Software Engineering, Web/Mobile Development, Databases, Cloud Applications, Machine Learning, Operating Systems, Networks, Security
- **Organizations:** Association for Computing Machinery (Officer), App Development Club, **5x Hackathon Winner**

Work Experience

Netflix June 2026 – September 2026
Software Engineer Intern *Los Gatos, CA*

- Spearheaded the CI test execution pipeline for Netflix's internal UI platform serving **300+ apps across 20+ teams**
- Created a Claude Code developer tool for 100+ authors to detect backend-mutating requests, scaffold E2E suites, and generate API mock contracts, protecting **60+ Tier-0/1 Commerce workflows** during a large-scale migration
- Drove team consensus on a required-but-bypassable merge gate by weighing 3 options in a design doc, then shipped it as a GitHub check run that **gated 200+ pull requests in its first 6 weeks**, linking traces and screenshots on failure
- Proved all 3 write modes (real, mocked, blocked) worked end to end by building and deploying a federated GraphQL CRUD backend in Java Spring Boot and Docker, giving the team **its first backend it could safely mutate**

Workiva June 2025 – September 2025
Software Engineer Intern *Ames, IA (Remote)*

- Delivered production fixes for 20+ defects across a Sustainability platform supporting **940+ enterprise clients**, resolving UI state, DataGrid filtering, and input validation issues to ensure accurate data collection
- Engineered AWS SQS dead-letter queues in Kotlin/Java Spring Boot, **preventing migration halts** on failing records
- Raised platform test coverage to **90%** by writing 100+ unit and functional tests with React Testing Library
- Streamlined how enterprise clients collect **thousands of auditable ESG data points** by shipping task delegation features with React, Redux, and Dart on the frontend and Kotlin, Java Spring Boot, and MySQL on the backend

Oregon State University – Center for Applied Systems and Software (CASS) May 2024 – June 2026
Software Developer *Corvallis, OR*

- **Led a team of 8** developers and researchers to build the Rx Product Market Orchestrator, a React platform with containerized Flask microservices, **presented at ASME 2024** for innovations in healthcare supply chain logistics
- Wrote the gRPC Flask API backend for a 4-microservice containerized medical supply recommendation platform with a React.js frontend and C# .NET 8 middleware, following Agile and Scrum practices in Azure DevOps
- Built a secure, scalable user authentication system using Azure AD B2C, MSAL, and OpenID Connect
- Increased system reliability by writing **50+ unit tests** for API endpoints in C# .NET 8 with Moq and Fluent Assertions

Projects

Real-Time Pose-Tracking Game | *React, MediaPipe, WebSockets, Cloudflare Workers* Demo | Repository

- **Placed 3rd** among 70+ teams and 250+ participants at QuackHacks with a real-time motion-controlled game
- Converted MediaPipe landmarks into joint-angle rotations for pose tracking that required **zero per-player calibration**
- Synced live multiplayer state via WebSockets on Cloudflare Workers for low-latency gameplay across both players

Sound-Direction Wearable | *Raspberry Pi, Python, Gemini, WebSockets* Demo | Repository

- **Won Best Use of Gemini** at BeaverHacks among 100+ teams with an assistive wearable for single-sided deafness
- Built a full-stack pipeline processing live 360-degree mic array data into direction-of-arrival estimates, streamed via WebSockets to a real-time HUD, with Gemini as an async reasoning layer flagging urgent or user-directed sounds

Smart Alarm App | *Swift, SwiftUI, watchOS, HealthKit, CoreMotion, Core Data* Live Site

- **Led a 3-person team** over a 9-month capstone to build an Apple Watch alarm app that wakes users during light sleep
- Classified sleep stages in real time from **millions of data points** of 50Hz motion and heart-rate sensor data
- Cut writes and battery drain via Core Data batch-saving, syncing Watch-to-iPhone data over WatchConnectivity

Movie Guessing Game Website | *React, TypeScript, Express, MongoDB, AWS* Demo | Repository

- **Won BeaverHacks Winter 2024** among 40+ participants with an Express.js and MongoDB movie trivia game
- Deployed the API service on AWS EC2, stored files in AWS S3, and used CloudFront CDN for quick response times

Technical Skills

Languages: Java, Kotlin, TypeScript, JavaScript, Python, C#, Swift, Go, Dart, C, C++, SQL, Bash, HTML, CSS, YAML
Frameworks: Spring Boot, Node.js, Express.js, Flask, FastAPI, ASP.NET Core 8, React Native, SwiftUI
Libraries: React, Redux (RTK Query), Playwright, React Testing Library, JUnit, PyTorch, MediaPipe
APIs & Practices: REST APIs, GraphQL, WebSockets, Microservices, CI/CD, Unit Testing, End-to-End (E2E) Testing
Databases: PostgreSQL, MongoDB, MySQL, SQLite, Firebase
Cloud & DevOps: AWS (EC2, S3, CloudFront, SQS), Azure (AD B2C, DevOps), Docker, Jenkins, GitHub Actions
Developer Tools: Git, GitHub, Linux, Postman, Jira, Xcode, IntelliJ IDEA
AI: Claude Code (skills, MCP), Claude API, Gemini API, LLMs, AI Agents, Codex, Cursor